

CSE321 Quiz 04 Set B**Time 20 Minutes**

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Q1. Consider a system with TLB miss ratio of 4.25%, and the main memory access time of 125 ns. If Effective Access Time is 142.3125 ns, then calculate the TLB access time. [2]

Q2. In a system there are 3 processes- **P1 (9 bytes)**, **P2 (6 bytes)** and **P3 (7 bytes)** with page size of 2 bytes. The size of the main memory is 32 bytes. Page tables of processes are given below.

Page Tables					
Process P1		Process P2		Process P3	
p	f	p	f	p	f
0	7	0	15	0	2
1	12	1	13	1	6
2	1	2	9	2	3
3	5			3	8
4	4				

Find corresponding physical addresses of the following logical addresses: [1.5+1.5]

- a. Address 01001 of P1
- b. Address 11011 of P3.

Q3. In a 16 bit Memory Address Architecture, the frame size is 1KB (2^{10} bytes). The paging System takes 8 bytes for each page entry. What is the maximum size of a process in this scenario? [3]

Q4. Consider this two scenario: [2]

- a. OS requests for page 5, where the page table has a total 4 page entries.
- b. OS requests for page 3, where the page table shows the valid bit is 0 (meaning invalid).

Now both of these situations are referred to as page fault, but the nature of them are different. Explain how each case is handled by the OS. (Each answer has to be within 3 lines or else it will be negatively marked.)